Experiment No. 5

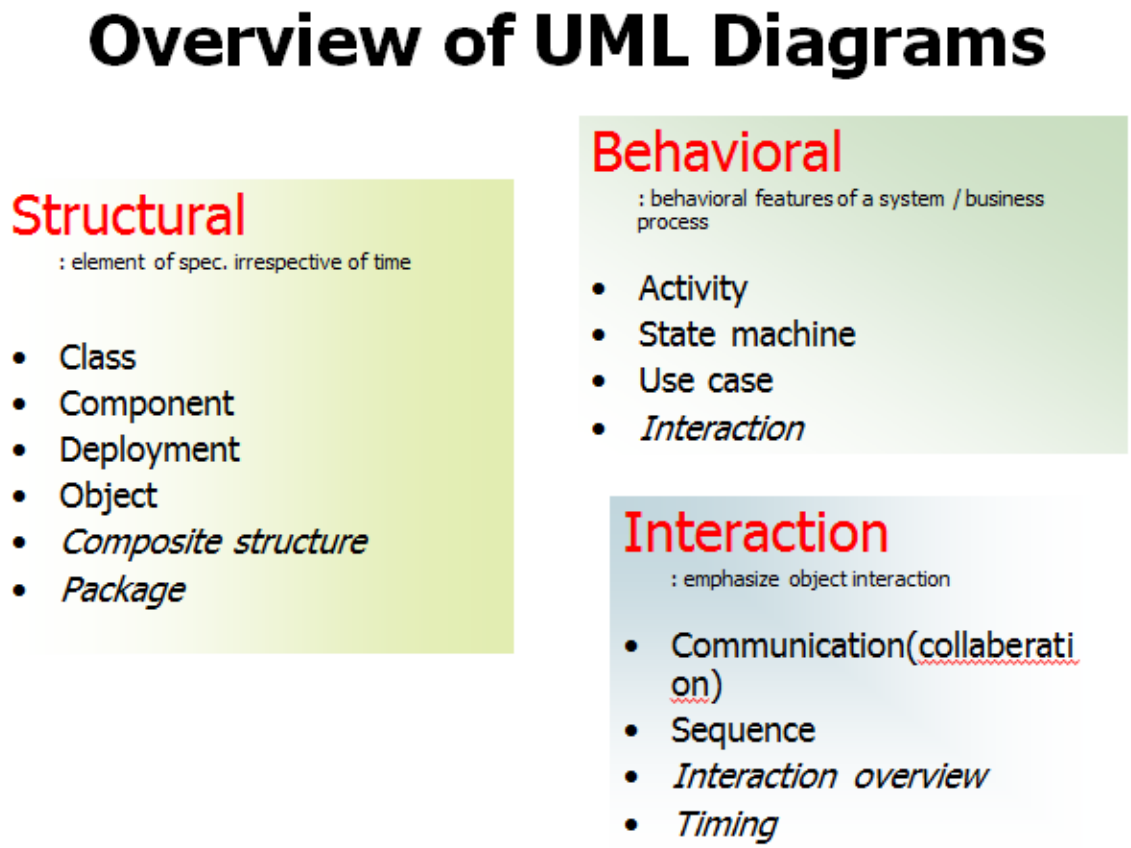
Title: Creation of Behavioral UML Diagrams

Batch: B3 **Roll No.:16014225801** **Experiment No.: 5**

Aim: Creation of Behavioral UML Diagrams

Resources needed: IBM Rational Rose/Open Source UML Tool-star UML

Theory



UML specification defines two major kinds of UML diagram: **structure diagrams and behaviour diagrams.**

Structure diagrams show the static structure of the system and it's parts on different abstraction and implementation levels and how they are related to each other. The elements in a structure diagram represent the meaningful concepts of a system, and may include abstract, real world and implementation concepts.

Behavior diagrams show the dynamic behavior of the objects in a system, which can be described as a series of changes to the system over time.

Behaviour Diagram:
Behavior diagrams show the dynamic behavior of the objects in a system, which can be described as a series of changes to the system over time.

1. Use case diagrams are behavior diagrams used to describe a set of actions (use cases) that some system or systems (subject) should or can perform in collaboration with one or more external users of the system (actors) to provide some observable and valuable results to the actors or other stakeholders of the system(s).

2. Interaction diagrams include several different types of diagrams:

- **Sequence diagram** is the most common kind of interaction diagrams, which focuses on the message interchange between lifelines (objects).
- **Interaction overview diagram** defines interactions through a variant of activity diagrams in a way that promotes overview of the control flow. Interaction overview diagrams focus on the overview of the flow of control where the nodes are interactions or interaction uses. The lifelines and the messages do not appear at this overview level.
- **Communication diagram (previously known as Collaboration Diagram)** is a kind of interaction diagram, which focuses on the interaction between lifelines where the architecture of the internal structure and how this corresponds with the message passing is central. The sequencing of messages is given through a sequence numbering scheme.
- **Timing diagrams** are used to show interactions when a primary purpose of the diagram is to reason about time. Timing diagrams focus on conditions changing within and among Lifelines along a linear time axis.

3. Collaboration diagram
A collaboration diagram is an interaction diagram that emphasizes the structural organization of the objects that send and receive messages.
We form a collaboration diagram by first placing the objects that participate in the interaction as they exist in a graph, and then add the links that connect these objects as arcs of this graph. Finally adorn these links with the messages that objects sends and receive with sequence numbers.

Need of collaboration Diagram:
We use collaboration diagram to describe a specific scenario. Numbered arrows show the movement of messages during the course of scenario. A distinguishing feature of a collaboration diagram is that it shows the objects and their association with other objects in the system apart from how they interact with each other. The association between objects is not represented in a sequence diagram.

Elements of collaboration Diagram:
A sophisticated modelling tool can easily convert a collaboration diagram into a sequence diagram and the vice versa. Hence, the elements of a Collaboration diagram are essentially

the same as that of a sequence diagram. 2 UML State machine diagram and activity diagram are both behavioural diagrams but have different emphases.

4. Activity diagram shows sequence and conditions for coordinating lower-level behaviors, rather than which classifiers own those behaviors. These are commonly called control flow and object flow models.

5. State machine diagram is used for modelling discrete behavior through finite state transitions. In addition to expressing the behavior of a part of the system, state machines can also be used to express the usage protocol of part of a system. These two kinds of state machines are referred to as behavioral state machines and protocol state machines.

Results:

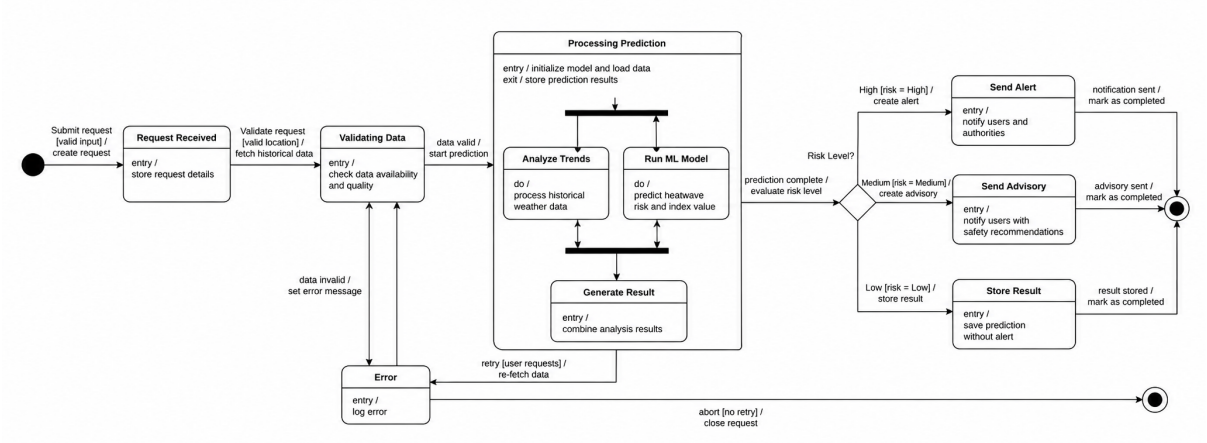


Fig. State Machine Diagram

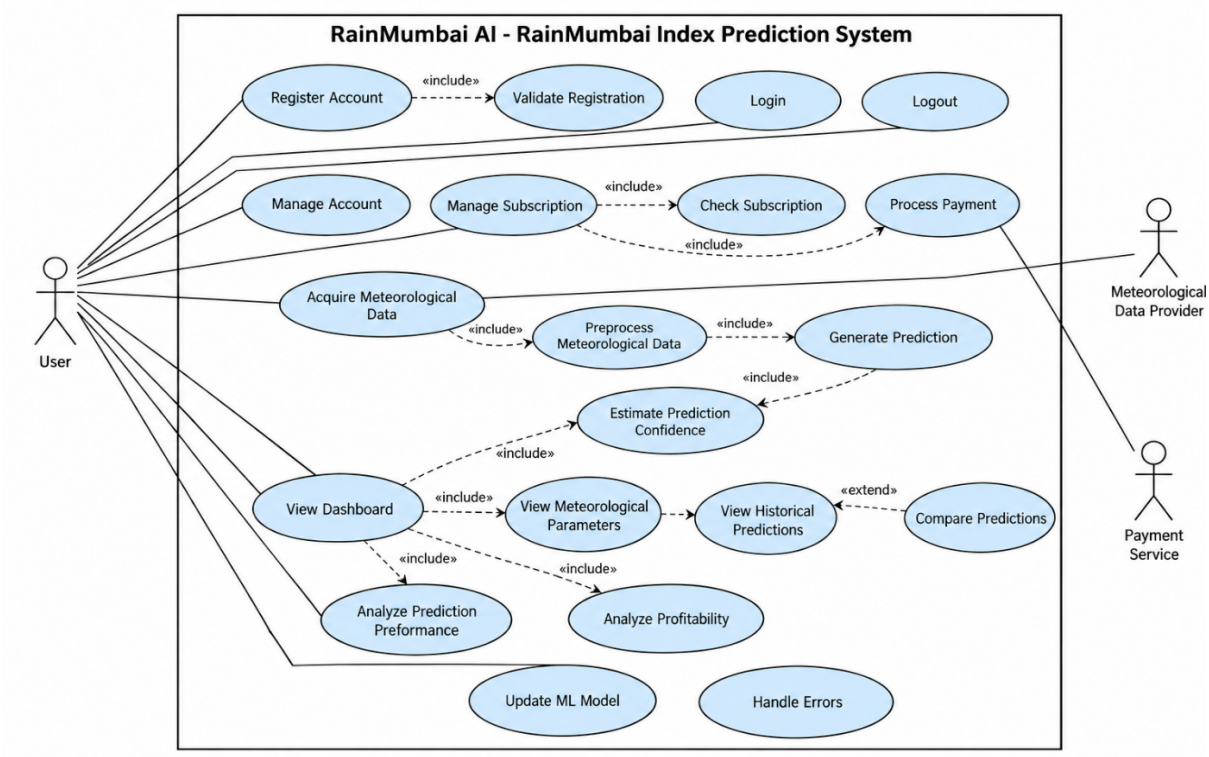


Fig. Use Case Diagram

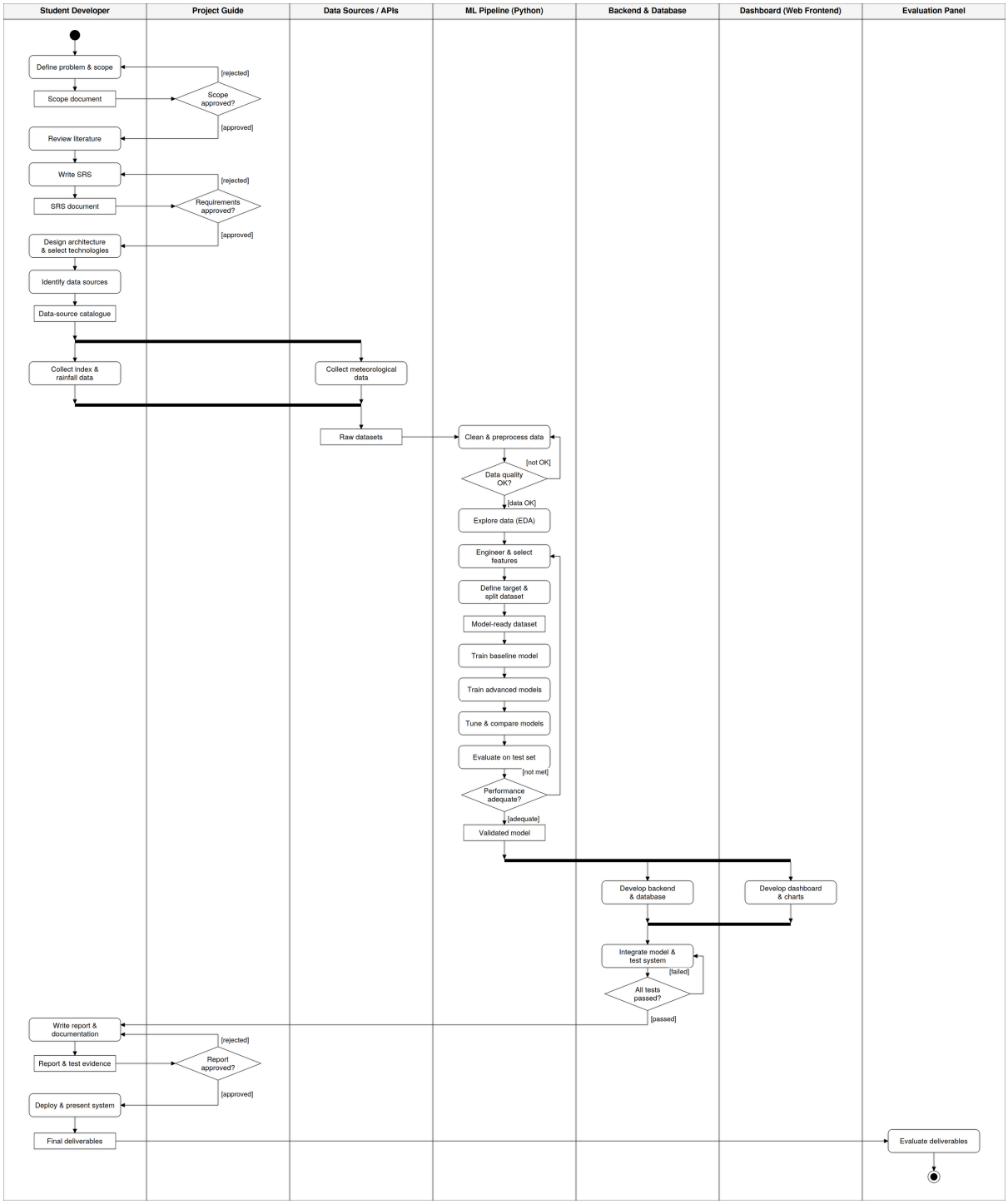


Fig. Activity Diagram

Questions:

1. In a use case diagram, relationships between different actors are normally shown.
2. A Communication diagram specifies a scenario.
3. Explain the difference between activity diagram and sequence diagram.

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1. In a Use Case Diagram, relationships between actors are not normally the main focus. The diagram mainly shows the interactions between actors and the system through use cases. However, relationships between actors can be shown using generalization, where one actor inherits the behavior or roles of another actor.

2. A Communication Diagram specifies a scenario by showing how objects or components interact with each other through messages. It focuses mainly on the relationships and communication links between objects, along with the sequence in which messages are exchanged to accomplish a particular task.

3. An Activity Diagram represents the workflow or sequence of activities in a system, focusing on actions, decisions, and the flow from one activity to another. A Sequence Diagram, on the other hand, represents interactions between objects over time, focusing on the messages exchanged between them and the order in which those messages occur. In simple terms, an activity diagram shows what activities are performed and how the process flows, while a sequence diagram shows which objects communicate and when they communicate.

Conclusion:

The experiment successfully demonstrated behavioral UML diagrams, including use case, activity, state machine, and collaboration diagrams, helping understand system behavior, workflows, interactions, and object communication.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

Books/ Website:

1.Michael Blaha, James Rumbaugh, “Object-Oriented Modeling and Design with UML”, Prentice-Hall of India, 2nd Edition

2. Mahesh P. Matha, “Object-Oriented Analysis and Design using UML”, Prentice-Hall of India

3. Timothy C Lethbridge, Robert Laganiere, “Object-Oriented Software Engineering – A practical software development using UML and Java”, Tata McGraw-Hill, New Delhi.

4. http://www.uml-diagrams.org/uml-23-diagrams.html